

Fundamentals of Programming for Business (CS2016)

Sessional 1 Exam

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 3

Date: February 23rd, 2026

Course Instructor(s)

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Roll No

BSCFE-2C

Section

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Student Signature

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Attempt all the questions.

CLO 2: Design and implement algorithms to solve real world business problems and should be able to translate a problem statement into pseudo-code/Python code;

Q1: E-Commerce Order Processing.

[8 Marks]

Consider the following program:

```
orders = [
    ("Ali", 5000, True), both
    ("Sara", 3000, False), none
    ("Hassan", 7000, True), both
    ("Zara", 2000, False) none
]
```

*name = names
amount = number
paid = True/False*

```
vip_customers = {"Ali", "Hassan"}
```

```
total_revenue = 0
```

```
discounted = []
```

```
for name, amount, paid in orders:
```

```
    if name in vip_customers and paid:
        amount *= 0.9
```

```
    if paid:
        total_revenue += amount
```

```
    else:
        discounted.append(name)
```

```
summary = {
    "total_orders": len(orders),
    "revenue": total_revenue,
    "pending": len(discounted)
}
```

```
print(summary)
```

```
print("Pending:", discounted)
```

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- (a) Predict the final output. Show all the working. (4 Marks)
(b) Explain why a set is used for vip_customers instead of a list. (2 Marks)
(c) If tuples were replaced by lists in orders, what would change? (2 Marks)

CLO 2: Design and implement algorithms to solve real world business problems and should be able to translate a problem statement into pseudo-code/Python code;

Q2: A company wants to automate its Employee Bonus System. [10 Marks]

The HR manager enters:

- Employee name
- Monthly salary
- Years of service
- Performance rating (1–5)

The bonus policy is:

1. If years of service ≥ 5 AND rating ≥ 4
→ Bonus = 20% of salary
2. If years of service ≥ 3 AND rating ≥ 3
→ Bonus = 10% of salary
3. Otherwise
→ Bonus = 5% of salary

Additionally:

- If salary is less than 30,000, print: "Entry Level Employee"
- Otherwise print: "Senior Employee"

Requirements:

a) Write the pseudocode for the problem. (6 marks)

b) Clearly identify: (4 marks)

- Input
- Processing
- Output

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CLO 3: Apply basic programming concepts

Q3:

[12 Marks]

Part A – Find the Error

(6 Marks)

Code 1

```
age = int(input("Enter age: "))
if age = 18:
    print("Eligible")
else:
    print("Not eligible")
```

1. Is there an error?
2. If yes, explain the error and correct it.

Code 2

```
score = int(input("Enter score: "))
if score >= 50:
    print("Pass")
else:
    print("Fail")
```

1. Identify the error.
2. Rewrite the correct version.

Part B – Dry Run

(6 Marks)

Carefully read the following code and write the exact output:

```
x = 15
y = 5
z = 0
```

```
if x > 10 and y < 10:
    z = x + y
else:
    z = x - y
```

```
if not z == 20:
    print("Value A:", z)
else:
    print("Value B:", z)
```

```
print("Done")
```

Show step-by-step evaluation of:

- First condition
- Second condition
- Final output